CEITEC

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# C9555: A survival guide to simulations

## Week 3

Denys Biriukov – Oct. 12th, 2023

# Acknowledgments

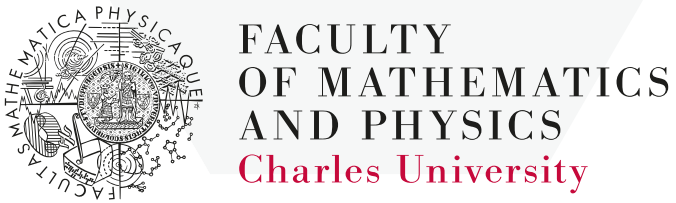
This part of the course is inspired and partially based on:

- Teaching materials of “Spring School in Computational Chemistry 2023” taken place in Finland and prepared by Dr. Hector Martinez-Seara and Dr. Matti Javanainen.
- Teaching course about molecular dynamics simulations at Charles University by prof. Lukasz Cwiklik.









# More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

# Get ready for practical session

- Wolf computers with GPU (more comfortable to work on)
  - wolf04 (not really good)
  - wolf05
  - wolf07
  - wolf08
  - wolf09
  - wolf12
  - wolf18
  - wolf19

# Get ready for practical session

- A bit of things to be done:
  - `mv ~/survival_guide_simulations_c9555/week02/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

# Get ready for practical session

- `git clone https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week03/`
- `bash ../infinity/infinity-temp.sh`

- `bash ../conda/conda-adjust.sh`
  - `bash ../conda/conda-light-activate.sh` (if does not work, first do this)
- `bash ../conda/jupyter-start.sh`
- `bash ../conda/conda-adjust-mda.sh`

# What is today?

- Week 3:
  Mixing Solvent – Physics and Chemistry in a Box.
- We prepared and run simulations with water/methanol mixtures.
- Today we will analyze them.
- Related theoretical concepts (touched the last week and today) will be discussed.

| TEACHING WEEK | DATE | LECTURER | TOPIC |
|---|---|---|---|
| 01 | 21.09 | TR | Getting Started (Linux, Python, VMD) |
| 02 | 05.10 | DB | Mixing Solvents – Physics and Chemistry in a Box – part I |
| 03 | 12.10 | DB | Mixing Solvents – Physics and Chemistry in a Box – part II |
| 04 | 19.10 | DB | Ions, Molecules, and Their Interactions |
| 05 | 26.10 | OK | Making Drugs *in silico* – part I – Lipid Membranes |
| 06 | 02.11 | OK | Making Drugs *in silico* – part II – Ligands Parametrizations |
| 07 | 09.11 | TR | Making Drugs *in silico* – part III – Steered Molecular Dynamics & Free Energy Calculations |
| 08 | 16.11 | TR | Making Drugs *in silico* – part IV – Free Energy Calculations & Permeability Coefficient |
| 09 | 23.11 | WM | How to Beat AlphaFold – part I |
| 10 | 30.11 | WM | How to Beat AlphaFold – part II |
| 11 | 07.12 | WM | Working in a Professional Environment |
| 12 | 14.12 | OK | Zápočet – first session to start your project |

# Today program

Lecture (1st hour)

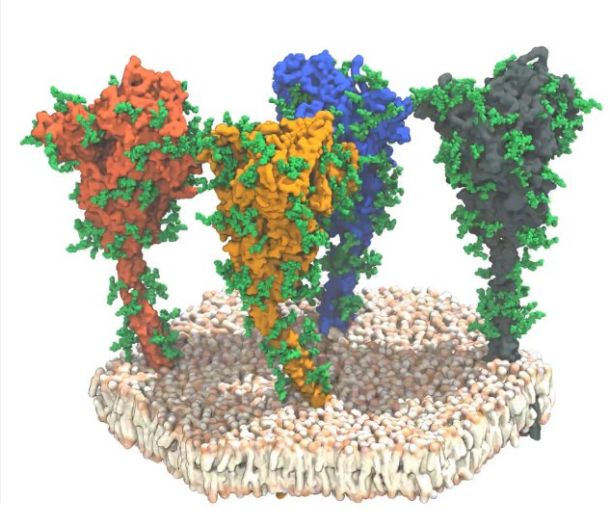
- Relation of molecular dynamics to thermodynamics.
- Ensembles in molecular simulations.
- Bridging simulation outputs and experimental observables.

Practical session (2nd and 3rd hours)

- Analysis of systems simulated in GROMACS.
- Convergence of simulation data and concepts of error estimation.
- Introduction to MDAnalysis as automatized and advanced post-processing.

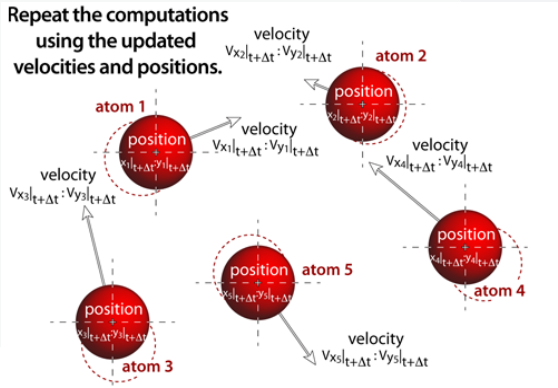
# Quick recall of the previous week

- Molecular dynamics simulations are based on physical laws and concepts.
- Atoms are represented as billiard balls that move according to Newton's equations of motion.
  - Born–Oppenheimer approximation.
- Force fields (predefined parameters) drive the evolution of the system.
- Time in simulation – time step, trajectories, equilibration, non-equilibrium systems.



Spike protein of SARS-CoV-2

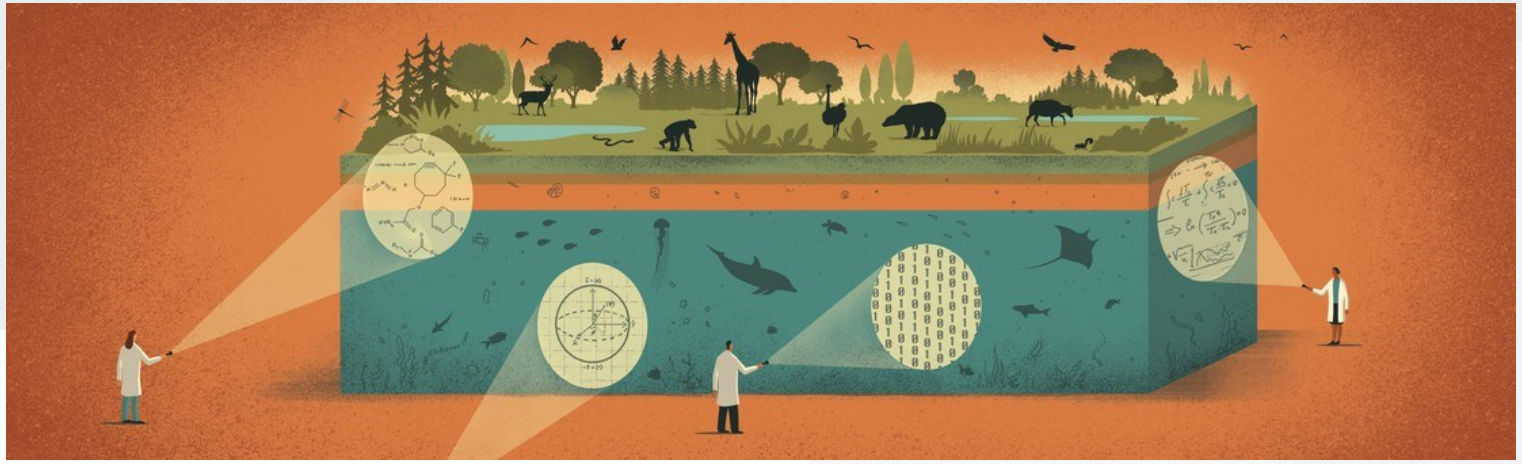
TUROŇOVÁ, SIKORA, SCHÜRMANN et al.
MAX PLANCK INSTITUTE OF BIOPHYSICS
PAUL-EHRLICH-INSTITUT




http://atomsinmotion.com/book/chapter5/md

# Bridging two worlds

- Simulations are a tool.
- We collect numbers.
- … but it is not the endgame.

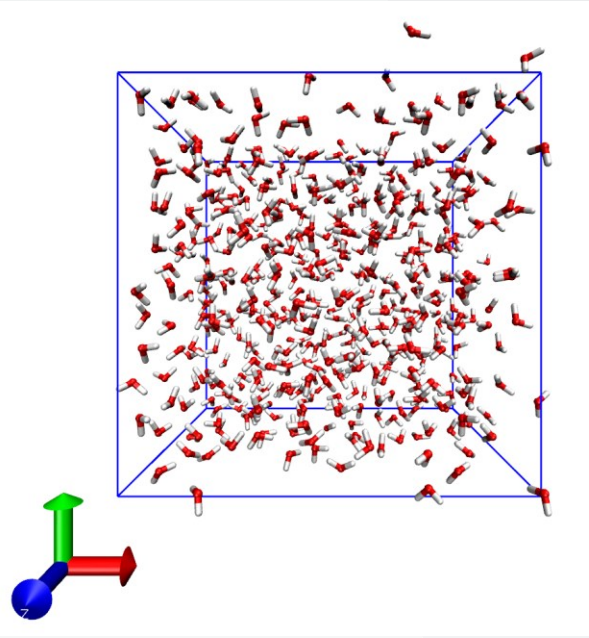


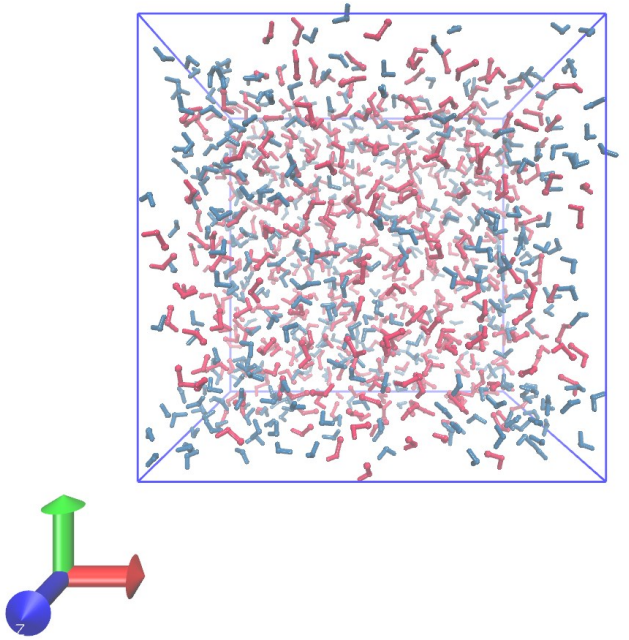
from https://elifesciences.org/subjects/physics-living-systems

- Massive amounts of numerical information should be transformed into answers to biological questions.
  - Simulation outputs compared to experimental observables.
- How to make this transformation?

# Micro and macro

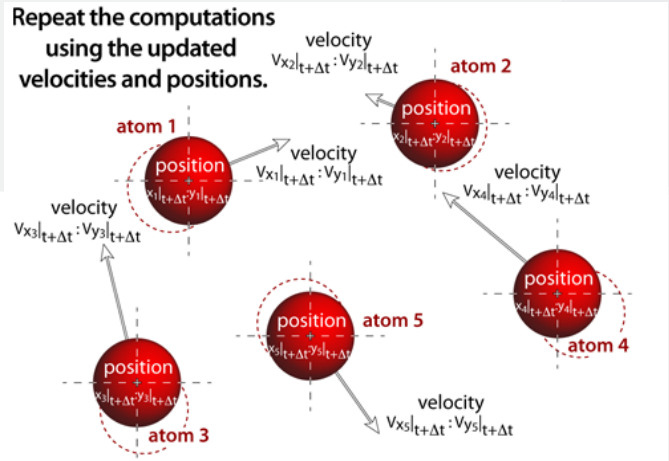
- Newton's equations of motion govern microscopic evolution (atomic positions, atomic velocities).
- They do not explicitly give macroscopic properties (e.g., temperature, pressure, etc.)
- Macroscopic properties are obtained from trajectories via statistical mechanics (statistical thermodynamics).



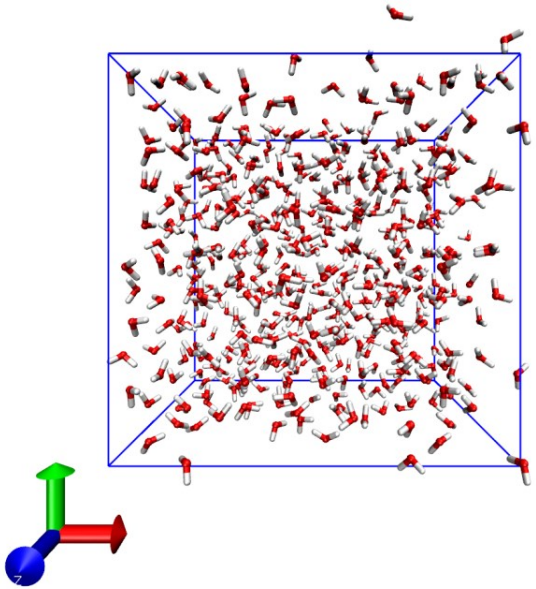


# Micro and macro

- Microscopic properties: $r\ (x, y, z)$, $v\ (v_x, v_y, v_z)$



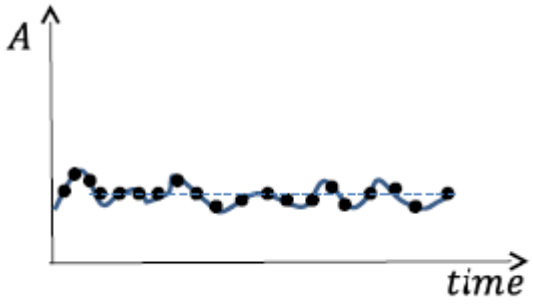

- Macroscopic properties: $T, P, V, \rho, \eta, \ldots$





# Ensembles and time

- Ergodic hypothesis: all accessible microstates are equiprobable over a long period of time.
  - Average over ensemble – average over time.
- Macroscopic properties are given by probability distribution within an ensemble.
- In molecular dynamics simulations, we “move” within the ensemble, reproduce its probability distribution.
  - Macroscopic properties can be estimated.
  - Applies to equilibrium molecular dynamics.

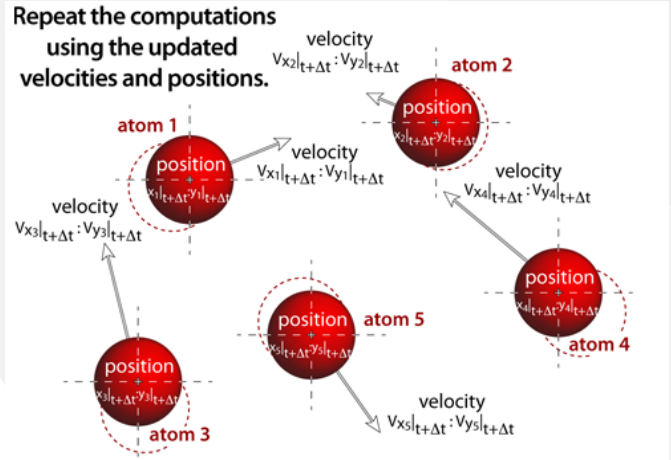



$$\langle A \rangle_{time} = \frac{1}{N} \sum_{i=1}^{N} A(t_i)$$

**under equilibrium**

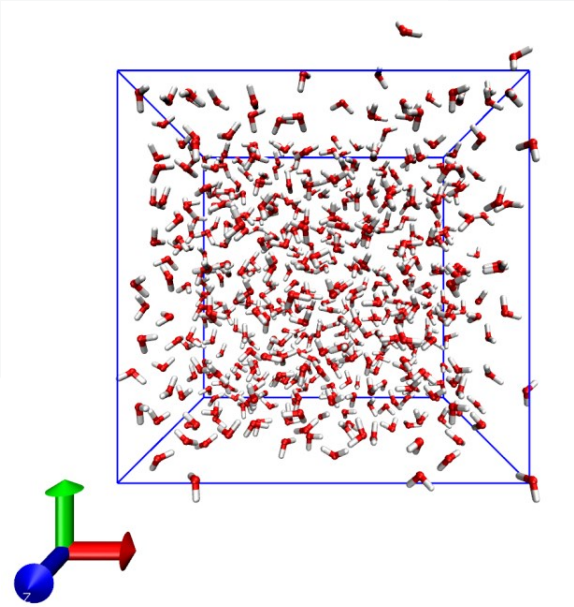
# Micro and macro

- If we have velocities, we can calculate kinetic energy and further, e.g., temperature and pressure.



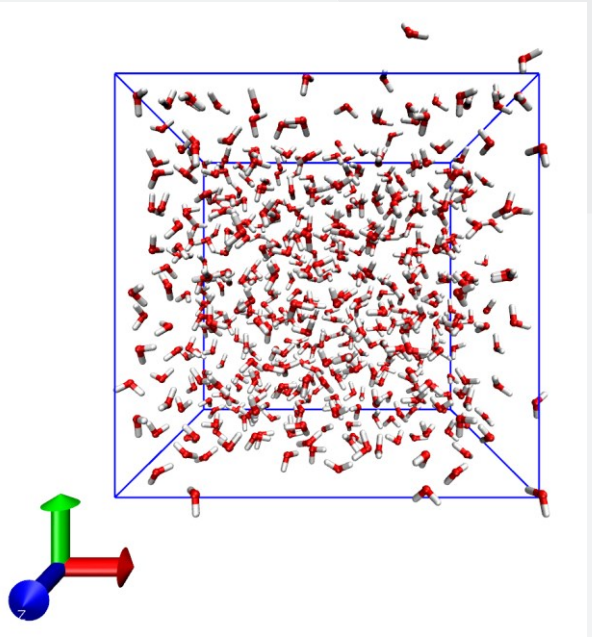

http://atomsinmotion.com/book/chapter5/md

- Relation to pressure:
  - $p = \frac{2N}{3V}\left\langle \frac{1}{2}mv^2 \right\rangle$
- Relation to temperature:
  - $T = \frac{2}{3k_b}\left\langle \frac{1}{2}mv^2 \right\rangle$, $k_b$ – Boltzmann constant



- However, Newton's equations of motion *per se* correspond to the constant energy in the system.
  - Temperature and pressure do not change unless they are adjusted by special algorithms .

# Statistical ensembles in MD



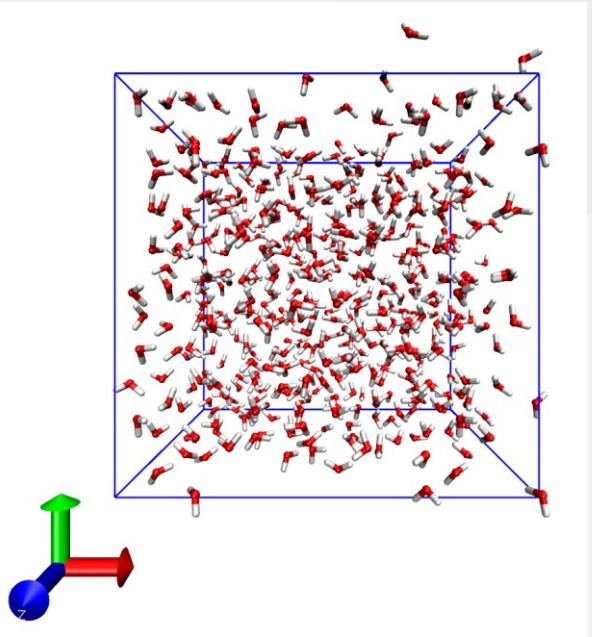
- Typically named by three letters.

- **NVE (microcanonical ensemble):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - E – energy (constant in a simulation).

- Energy is not added nor removed, fixed volume, particles just move in a locked box.
  - Newton's equations correspond to NVE (no external work, constant total energy, no heat exchange).

- Not used in practice: biological simulations are not isolated, big temperature fluctuations, not flexible, convergence and stability issues, etc.

# Statistical ensembles in MD



- **NVT (canonical ensemble, isothermal, Helmholtz canonical):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - T – temperature (constant in a simulation).
- Temperature is maintained at a desired value through the exchange of energy with the surroundings (conceptually, of course).
- Sometimes used in bio: simple, stable, realistic enough, excellent for first equilibration stages.
- Disadvantages – volume needs to change in many biosystems, compressibility pressure fluctuations, etc.

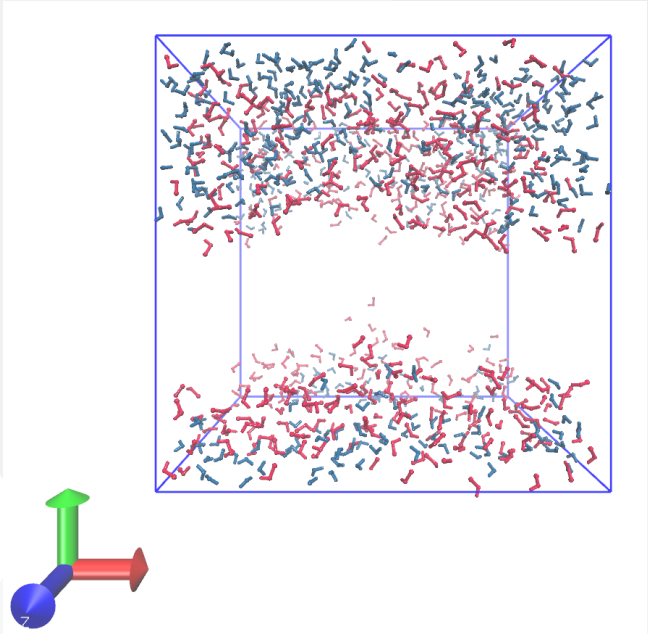
# Statistical ensembles in MD

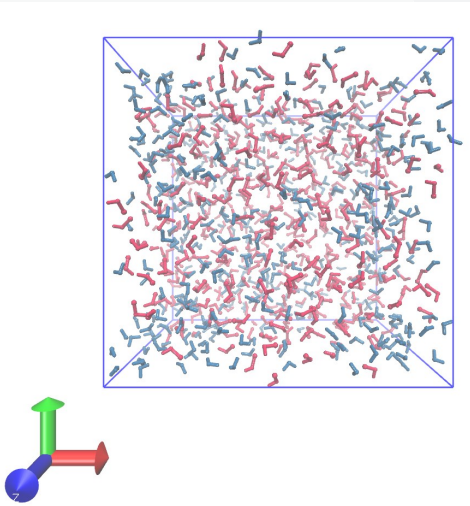
- **NPT (Gibbs canonical ensemble, isobaric-isothermal):**
  - N – number of particles (constant in a simulation).
  - P – pressure (constant in a simulation).
  - T – temperature (constant in a simulation).

- Both temperature and pressure are controlled; pressure is controlled through changing box dimensions.

- 99% use-case in biosimulations, replicates biological conditions, stable, flexible, direct comparison to experimental conditions.

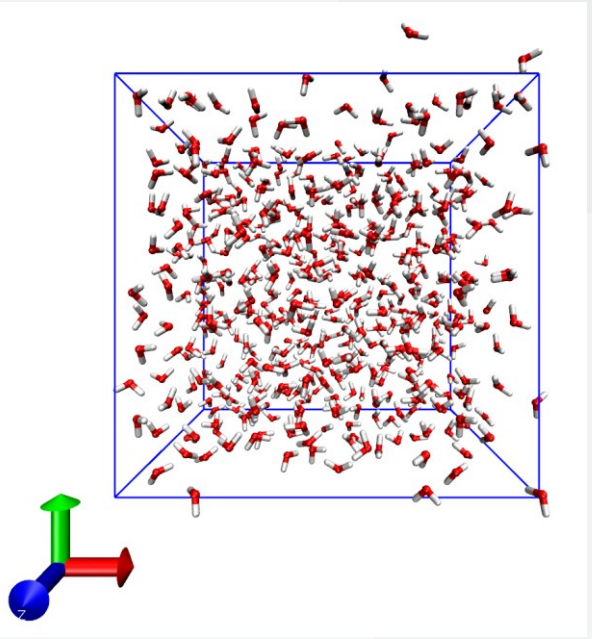
- Disadvantages – basically, technical and computational.





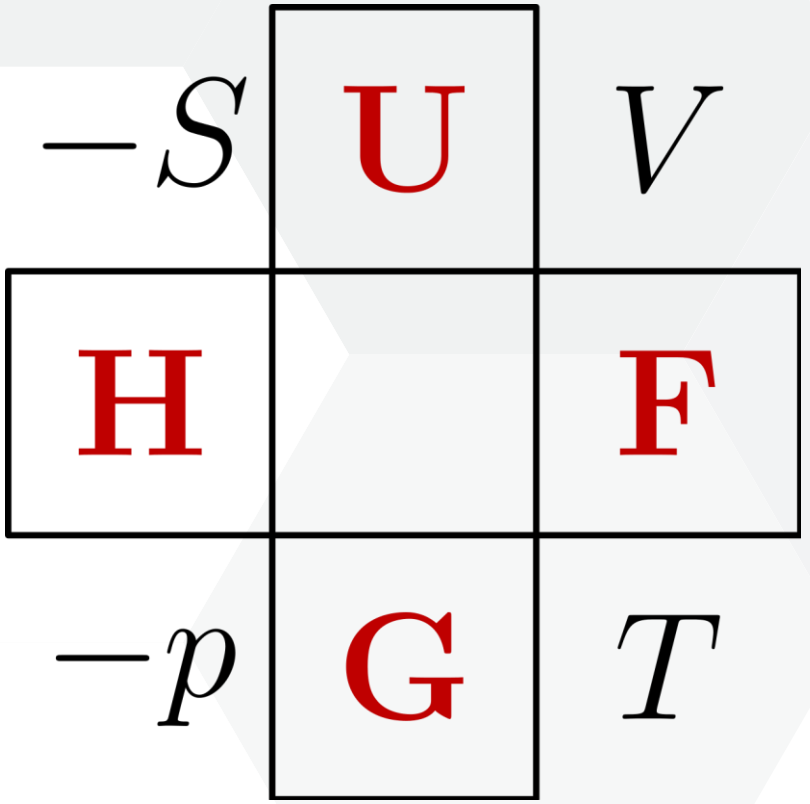
# Statistical ensembles in MD

- Other examples:
  - NPH (isoenthalpic-isobaric: number of particles, pressure, enthalpy).
  - NPE (isobaric: number of particles, pressure, energy).
  - μVT (Grand canonical: chemical potential, volume, temperature).
  - μPT (chemical potential, pressure, temperature).
    - μVT and μPT are usually used in Monte-Carlo simulations.
  - NPγT (number of particles, pressure, surface tension, temperature).
  - …
- I think you will never use them.



# Maxwell relations

- A set of equations in thermodynamics.
- Derived from the symmetry of second derivatives (complicated, I know).
- In simple words, these are equations that show how changes in one property (for instance, temperature) may affect another one (for instance, volume) under certain conditions.



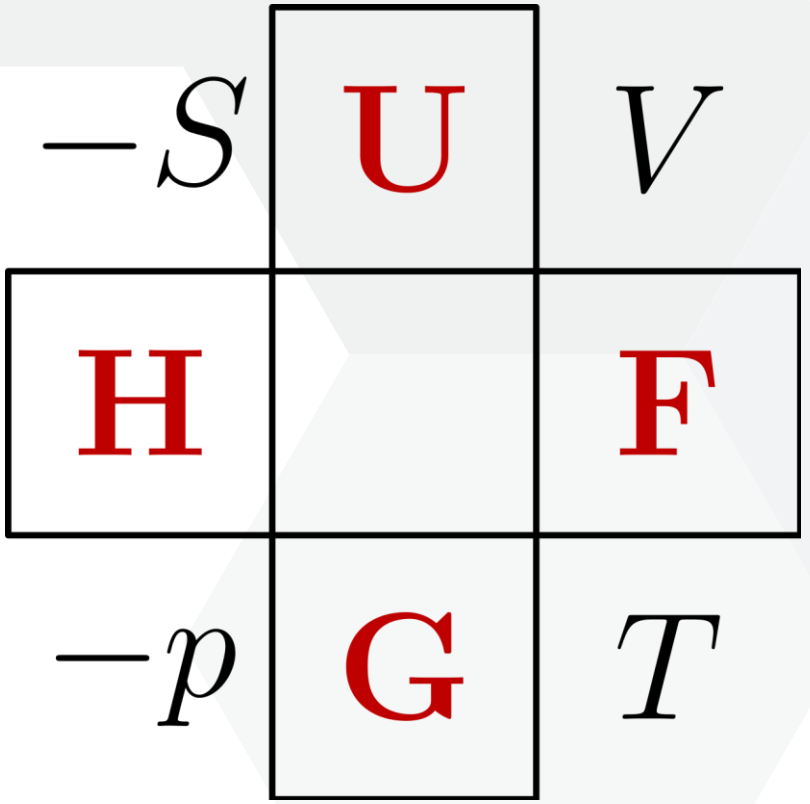

**Thermodynamic square**
picture from Wikipedia

# Maxwell relations

- p – pressure
- V – volume
- T – temperature

- S – entropy

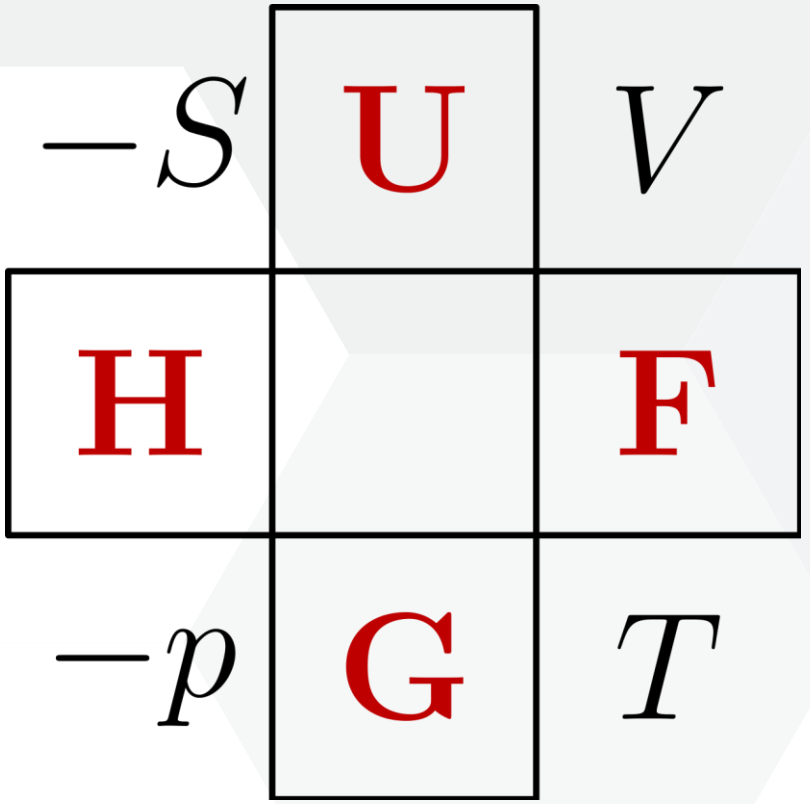
- U – internal energy
- H – enthalpy
- G – Gibbs free energy
- F – Helmholtz free energy (sometimes A)




**Thermodynamic square**
picture from Wikipedia

# Maxwell relations

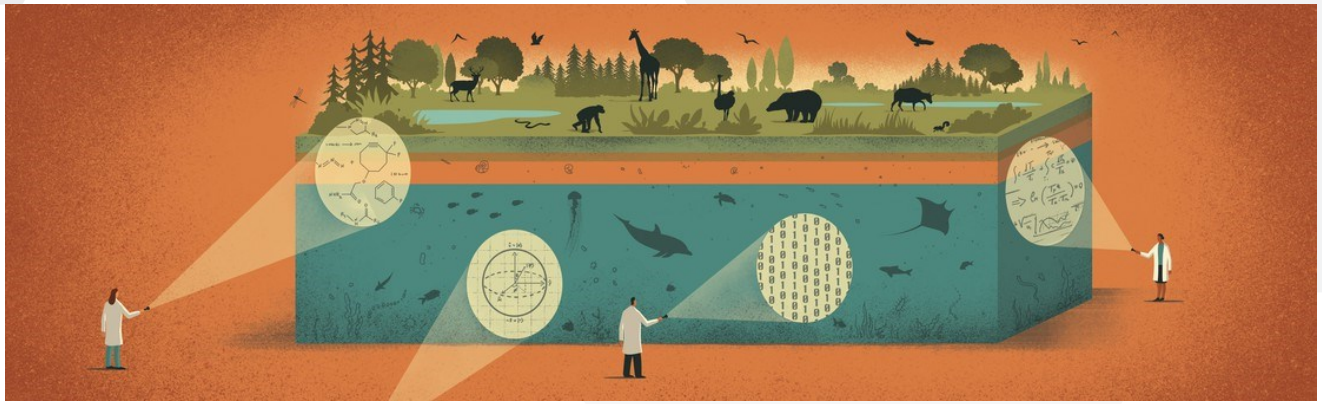
- U – internal energy
  - the total energy within a system
- H – enthalpy
  - energy of a system plus adsorbed/released heat at constant pressure
- G – Gibbs free energy
  - energy available to do work at constant temperature and pressure
- F – Helmholtz free energy
  - energy available to do work at constant temperature and volume




**Thermodynamic square**
picture from Wikipedia

# Simulations coupled to experiments

- The beauty of simulations is when they are linked to experiments.
- Compare your results to experiments whenever possible.
  - Synergy, not competition.
- We are scientists, not film makers.
  - Produce science, not sci-fi.
- “Garbage in, garbage out”.
- “Good in, good out”?
  - More delicate concept to think about.



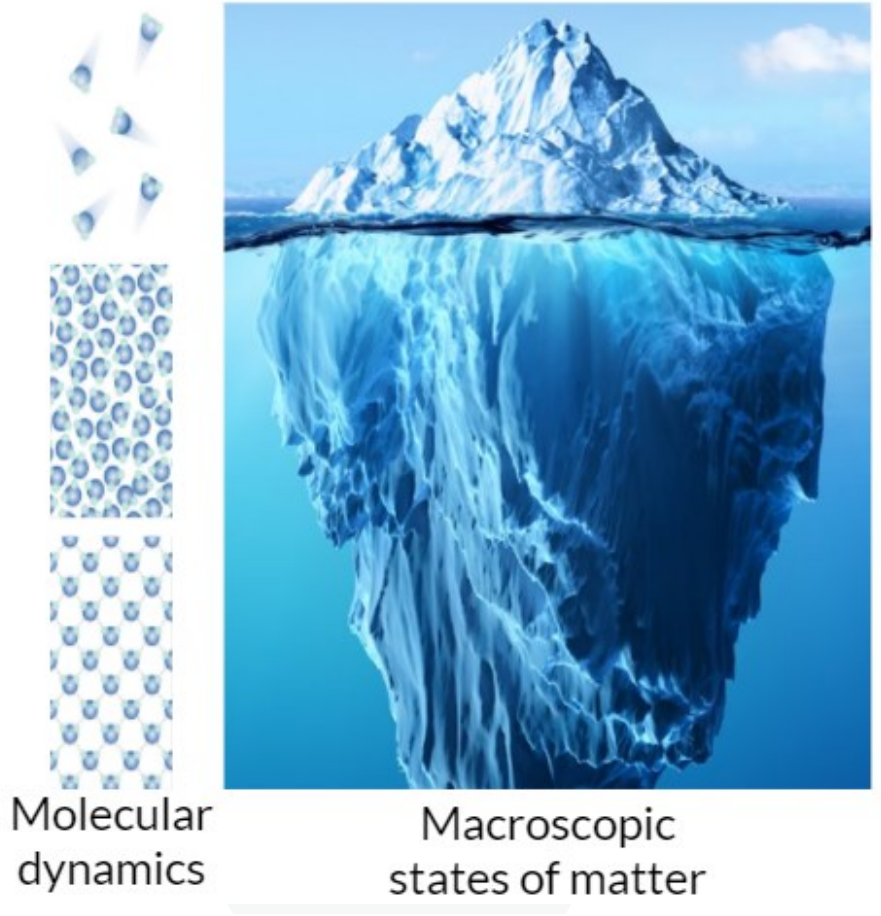
from https://elifesciences.org/subjects/physics-living-systems

# Properties to calculate and analyze

- Macroscopic properties we already discussed:
  - Temperature, pressure, volume, energy (basically, must-do)
  - Surface tension, polarization, etc.

- Microscopic properties:
  - Atomic and molecular distribution via radial distribution functions
  - Ordering – orientation, dipoles, …
  - Conformations – secondary structure of proteins, biological assemblies, …
  - Interactions – hydrogen bonds, adsorption, ligand binding, selectivity, …
  - Motion – diffusion, translocation, opening/closing of molecular gates, …

- Collected quantities can be categorized, for instance, structural or dynamic (e.g., self-diffusion coefficient).
  - In today practical.



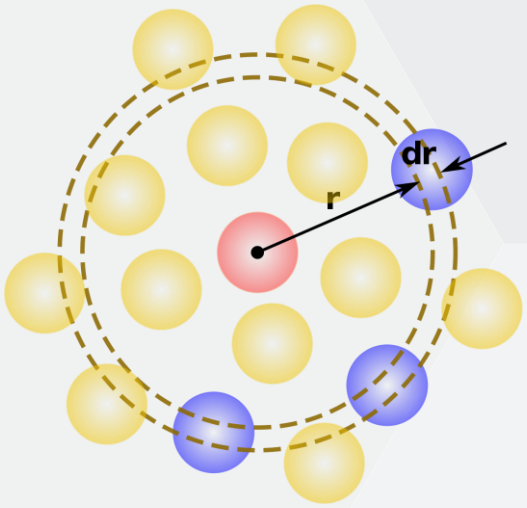

found on Twitter https://twitter.com/TrangAnhEstelle

# Radial distribution function (RDF, $g(r)$)

- A measure of the probability to find a particle at a distance $r$ from another reference particle.

- One of the pillars in post-processing MD simulations.
  - The easiest way to get the first “feeling” of what is happening in your simulations.
  - `gmx rdf`

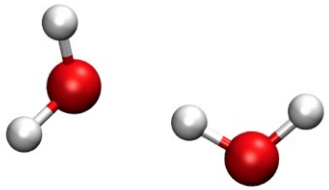
- RDF is related to the structure factor (via Fourier transformation), which can be determined from X-ray or neutron diffraction experiments.
  - One to one comparison of simulations and experiments.




picture from Wikipedia

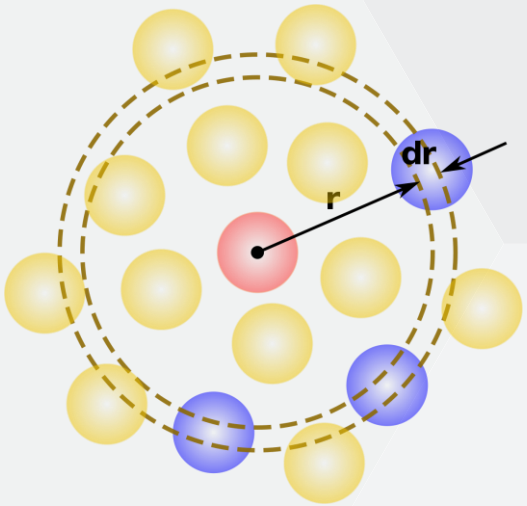
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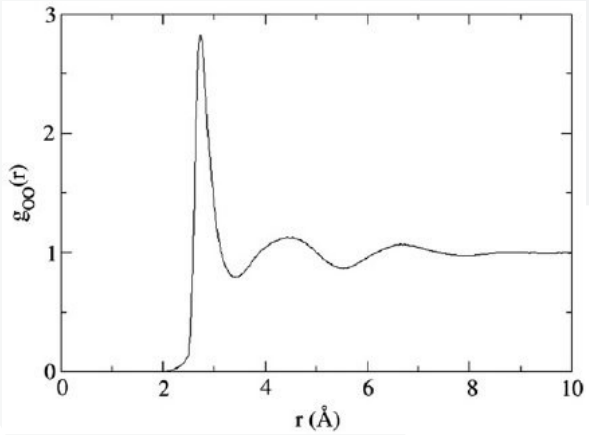


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picture from Wikipedia

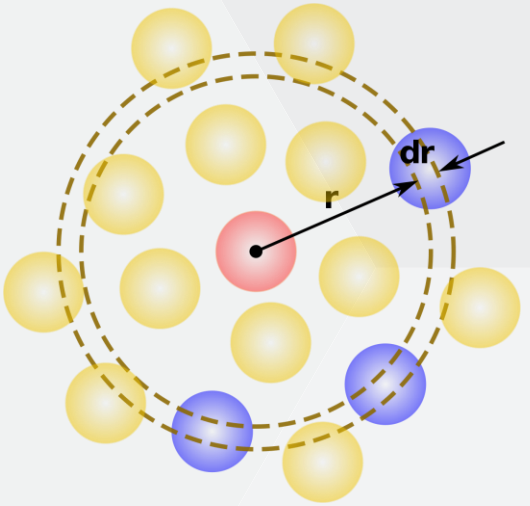



RDF of water oxygens around water oxygens from X-ray scattering data

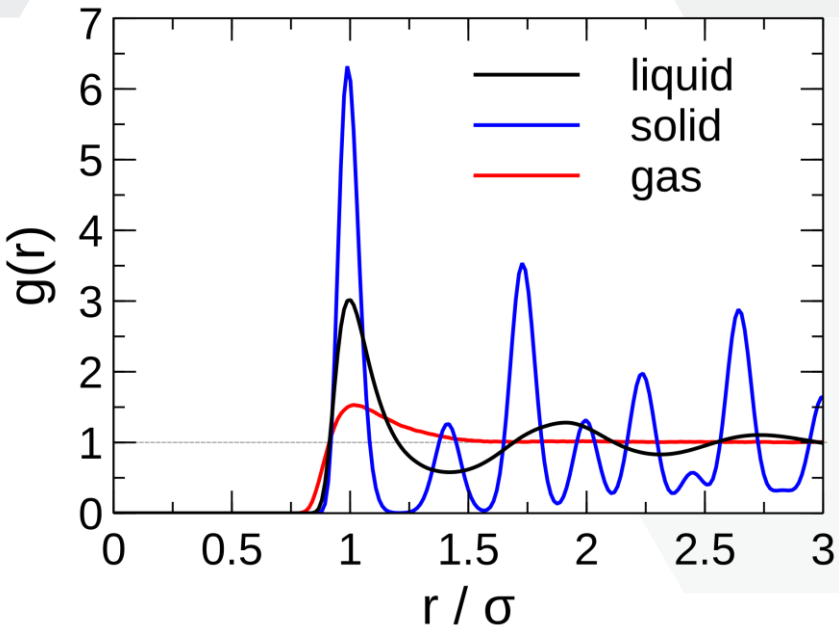
Clark et al. *Mol. Phys.* 2010, 108, 1415–1433

# Radial distribution function (RDF, $g(r)$)

- Defined peaks? More ordering!
- Larger peaks? More ordering!
  - Do not confuse height, width, and area.
- In most cases (especially biosimulations), RDFs converges to the value of 1.
  - Value that does not equal to one means that there is correlation
  - Correlations are typically gone at long distances (few nanometers max).




picture from Wikipedia

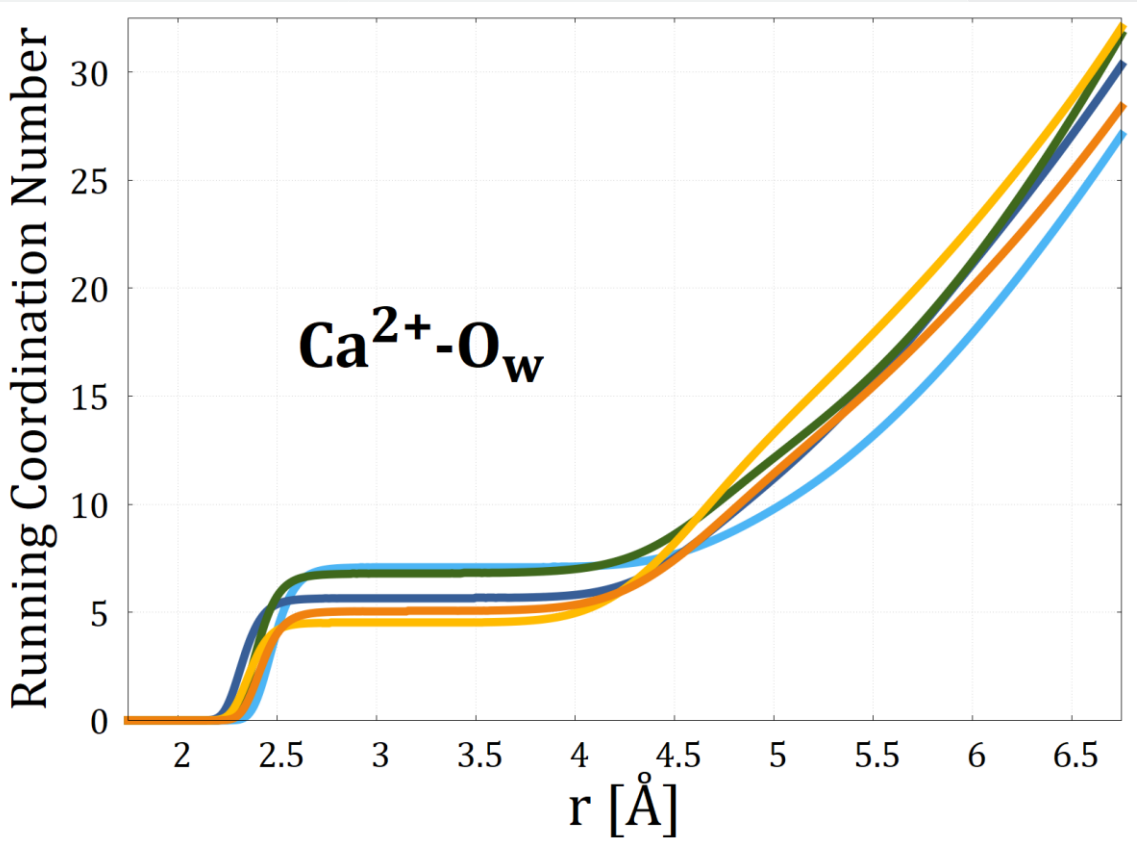



The radial distribution functions of solid (T = 50 K), liquid (T = 80 K), and gaseous argon (T = 300 K)

taken from Wikipedia

# Running coordination number

- Coordination number is essentially the running integral of RDF.
  - You know what an integral is, right?
- Quick way to estimate how many atoms/molecules are around reference particle at a certain distance.
  - Useful for defined structures such a hydration shell of an ion, ligand, could be even protein.

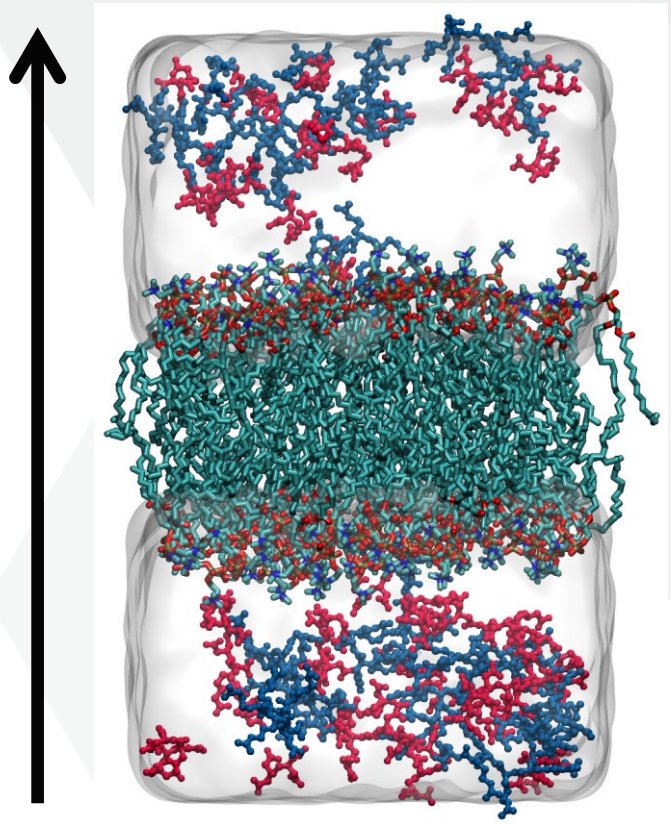



Running coordination number of water oxygens around calcium cation

Biriukov et al. *J. Chem. Phys.* 2022, 156, 194505

# Density distribution (number/mass/charge density)

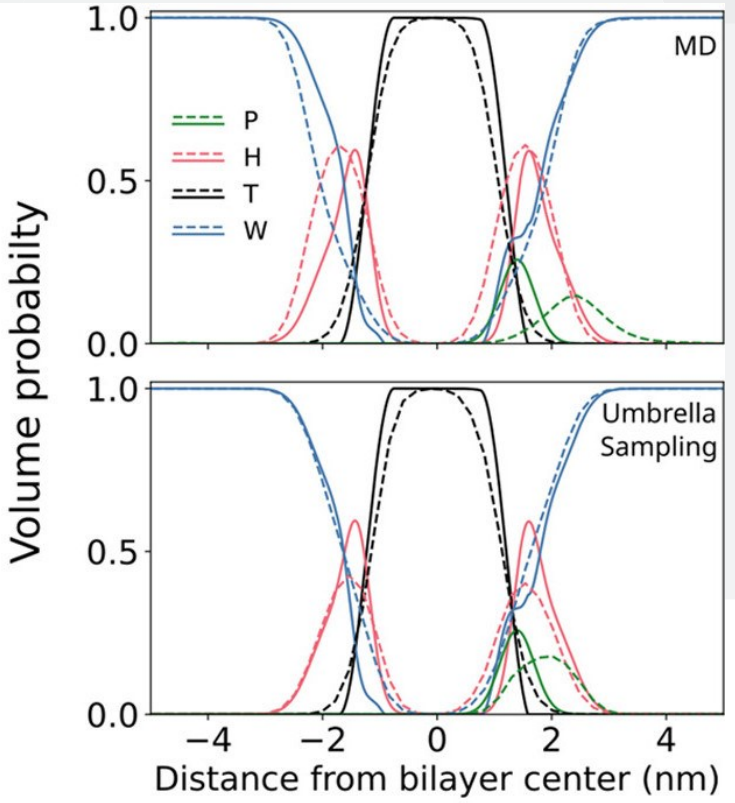
- Similar to RDF: distribution of particles but typically as a function of box dimension(s).
  - One-dimensional, lateral, three-dimensional, …
- Another pillar in post-processing MD simulations.
  - Especially relevant for interfacial systems. (membrane/aqueous interface, solid materials, water/air interface, etc.)
  - Inherently can possess information about orientation.
  - `gmx density`



Example of lipid bilayer system: density along the arrow would give you a lot of info

# Density distribution (number/mass/charge density)

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  - One-dimensional, lateral, three-dimensional, …
- Another pillar in post-processing MD simulations.
  - Especially relevant for interfacial systems. (membrane/aqueous interface, solid materials, water/air interface, etc.)
  - Inherently can possess information about orientation.
  - `gmx density`
- Density can be also extracted from X-ray techniques and compared to simulations.
  - SAXS – small-angle X-ray scattering (see right).

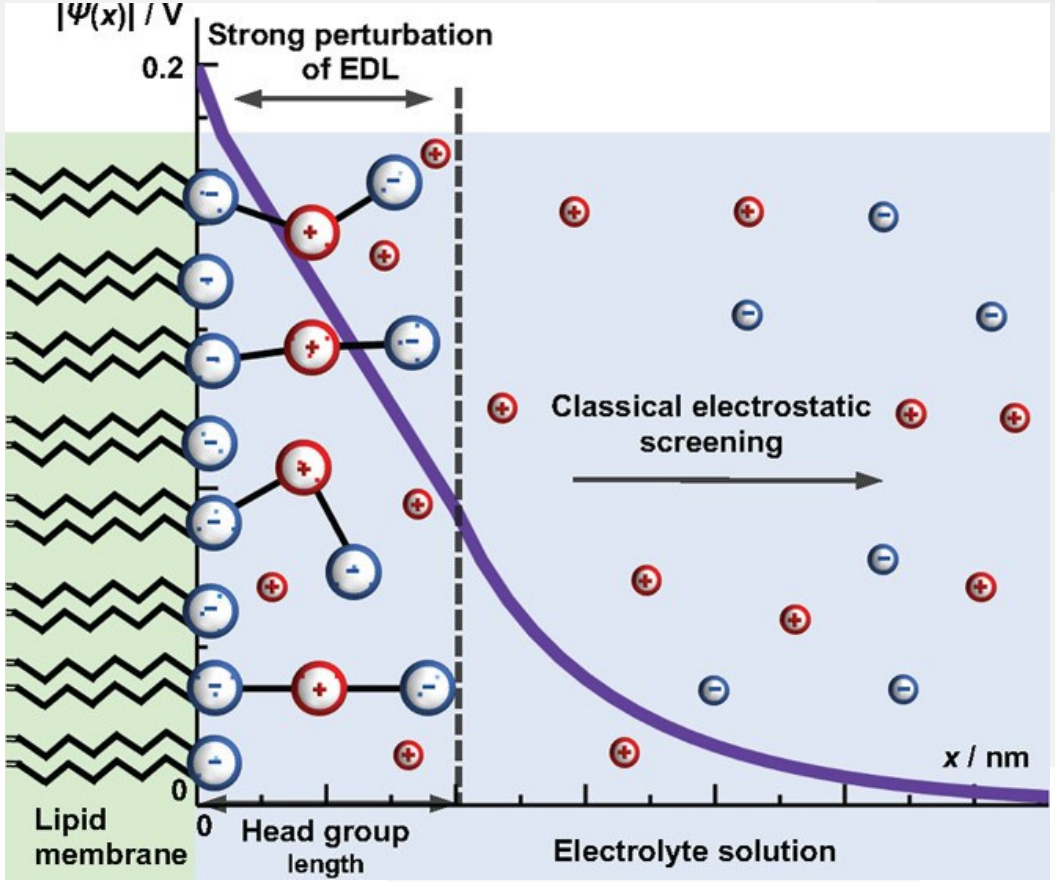



Density profile from simulations (dashed line) and SAXS experiments (solid line)

Carrer et al. *J. Phys. Chem. Lett.* 2023, 14, 7014–7019

# Angular distributions

- Information that is not that easily accessible by experiments.
  - Good for people doing simulations!
- Simple to define, various information to obtain.
  - Protein/peptide/drug orientation when adsorbed to the membrane or translocated through it.
  - Response of lipid headgroups to these processes (see right).
  - `gmx angle`

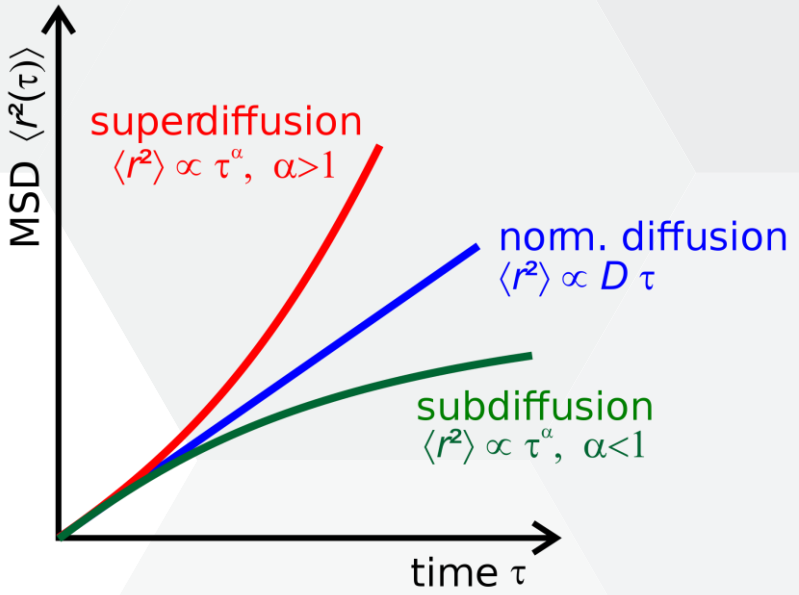



Schematic representation of lipid membrane/aqueous interface

Bohinc et al. *J. Chem. Theory Comput.* 2021, 18, 448–460

# Self-diffusion coefficient

- Self-diffusion coefficient:
  - A transport property.
  - Quantifies how rapidly particles move in a medium.
  - Indicator of molecular mobility.
  - Related to viscosity, conductivity, structural properties, etc.
- Typically derived from the Einstein relation.
  - There are details though (see right, more during practical).
  - `gmx msd`
- Important fact: sensitive to the box size in your simulations.
  - Is it true? We will check it today.
  - Why is it happening? We will learn next week.




Comparison of normal and anomalous diffusions

taken from Wikipedia

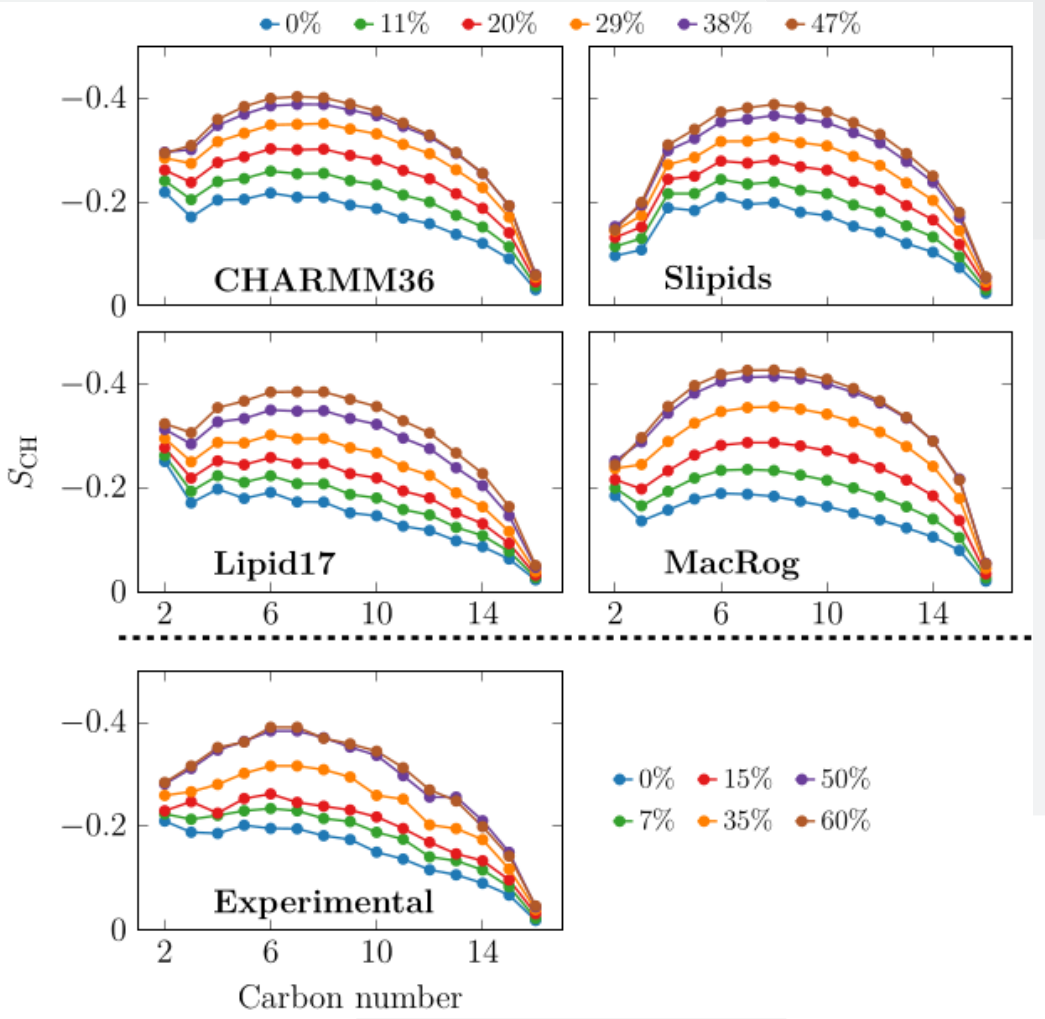
# Other measures (not today)

- Order parameters of lipid membranes.
  - Alignment of lipid tails (orientation of C–H bond) with respect to the normal (angle $\theta$).
  - Represents the ordering and packing of lipids.
  - Can be directly measured by both NMR and molecular simulations.
  - Highly relevant to describe multiple biological processes in cellular membranes.

- Commonly calculated as the deuterium order parameter.
  - $S = \langle 3\cos^2\theta - 1 \rangle / 2$
  - `gmx order`
  - Week 05.



**Ondřej Kroutil**




Order parameters for different lipid membranes with varying amount of cholesterol

Javanainen et al. *J. Chem. Theory Comput.* 2023, 19, 6342–6352

# Other measures (not today)

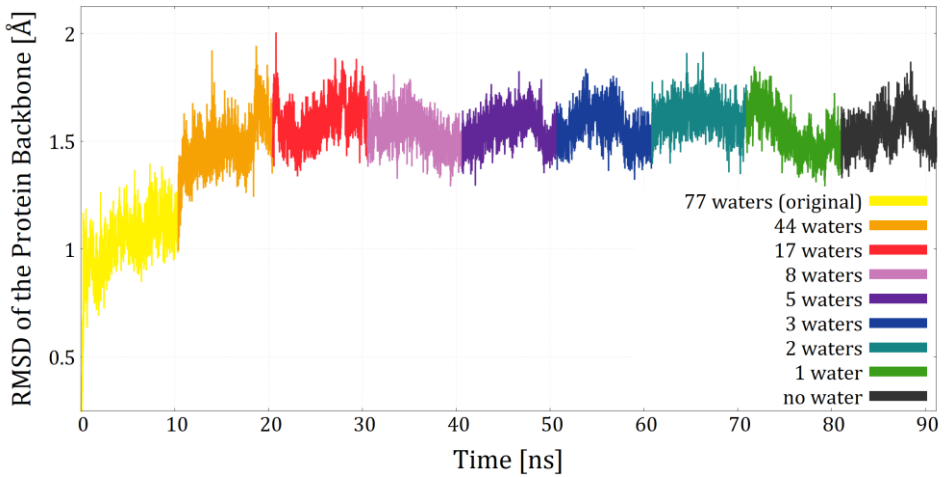
- Root-mean-square deviation (RMSD).
  - Must-do in simulations of proteins.
  - Excellent for equilibration monitoring and checking the stability of a protein structure.
  - `gmx rms`



Alignment of two proteins with a difference in a single mutation

taken from CHARMM-GUI

- Mutations/ligand binding can and usually do affect the RMSD of a protein.
  - If nothing is happening in your system, RMSD should become stable.




a typical RMSD graph

- `gmx dssp` – another tool to determine the secondary structure .

- Week 09.



**Willian Morton**

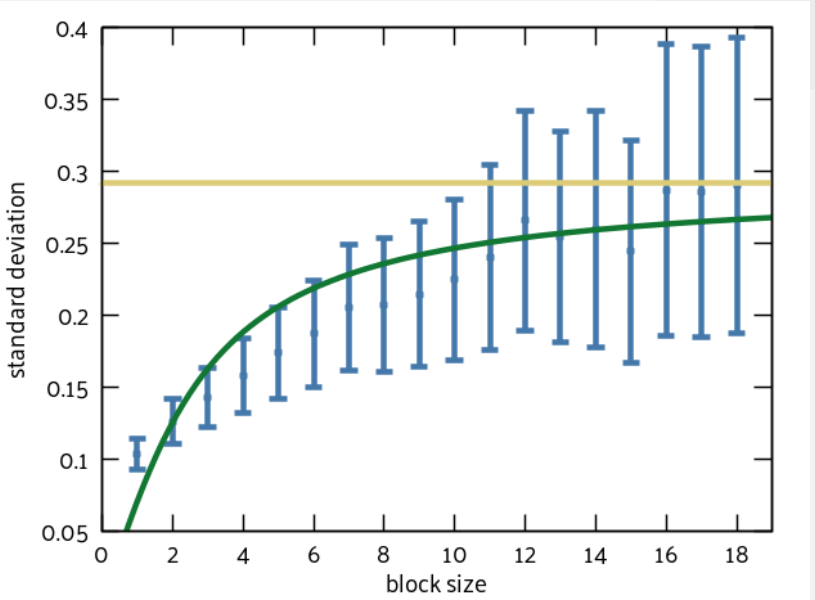
Kontkanen et al. *J. Chem. Phys.* 2022, 156, 175101

# Error analysis

- Whatever you report or even if you think your simulation is fantastic – error estimates are the must.

- Simulations are well-suited for reporting errors.
  - We calculate averages.
  - Time correlations.
  - Error of a mean.

- Standard deviation/error from:
  - Replica simulations = multiple simulations.
  - Divide one simulation in pieces.
    - In principle, good, but not ideal.
  - Block averaging.
    - `gmx analyze`




Example of block averaging: standard deviation depends on the size of the block

https://realerthinks.com/block-averaging-bootstrapping-estimating-mean-autocorrelated-data/

# Break!

# Get ready for practical session!